Predictory Project Report

Team Name: CHAT GPT
Case Study Chosen: 8 — AI for Inclusive MSME Growth
Project Name: Predictory — AI-Powered Bakery Intelligence Platform

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This document is prepared for the **technical team submission track**. The emphasis is on system architecture, implementation scope, engineering decisions, validation evidence, and technical business value. Product framing and interface decisions are included only where they help explain the implemented system and its operator workflow.

Introduction

Background of the Case Study

Predictory is a bakery operations decision-support system designed to empower businesses of all sizes, from independent single-shop bakeries to large multi-outlet chains. The project is grounded in a common operational reality: many bakery businesses already have sales data, inventory records, and basic reports, but next-day prep decisions are still made manually. In practice, this often leads to repeated overproduction of slow-moving baked goods, stockouts during peak periods, inefficient production allocation, excess ingredient purchasing, and inconsistent freshness across one or many locations.

The repository, seeded demo data, and product requirement documents position Predictory around a flexible operating model: while the prototype uses Roti Lane Bakery—a Malaysia bakery-cafe chain with a central kitchen and several outlets—as a comprehensive case study, the core logic is equally effective for a single-storefront operation. Whether managing a standalone shop's daily bake or a network of five seeded outlets with multiple SKUs, Predictory identifies realistic waste and stockout patterns. Rather than acting as a replacement for POS or ERP systems, Predictory is designed as a decision layer that helps operators decide what to prep, what to replenish, and where risk is likely to appear before service begins.

Why This Problem Matters: Statistics With Citations

The food-waste problem is large enough to justify targeted operational tools rather than generic reporting dashboards:

- Globally, **13.2% of food is lost after harvest and before retail**, and **19% more is wasted at retail, food-service, and household levels**, according to the latest FAO and UNEP triennial indicators [1].
- FAO also states that food loss and waste account for an estimated **8% to 10% of global greenhouse gas emissions**, linking the issue directly to climate and sustainability outcomes [1].
- In Malaysia, 2025 statistics cited by Deputy Prime Minister Datuk Seri Dr Ahmad Zahid Hamidi indicate the country disposes of **8.3 million tonnes of food waste annually**, with **24% of it still perfectly suitable for consumption**. This amounts to an alarming **260 kilograms of food waste per individual per year** [2].
- Furthermore, 2025 data from the Solid Waste and Public Cleansing Management Corporation (SWCorp) shows that total solid waste generation rose by over 10% in 2025, driven by economic activity and consumption changes, keeping food waste as the dominant component of landfills (often exceeding 30% to 40% during peak seasons like Ramadan) [3].

These figures support a strong sustainability case for better production planning in food-service operations, especially for perishable bakery products with short freshness windows.

Chosen SDG

The primary Sustainable Development Goal addressed by Predictory is **SDG 12: Responsible Consumption and Production**. The strongest alignment is through waste reduction, better use of ingredients, more disciplined production planning, and improved operational visibility for perishable goods. By helping bakery teams prepare closer to actual demand, Predictory aims to reduce avoidable waste without sacrificing service availability.

Predictory also supports **SDG 9: Industry, Innovation and Infrastructure** as a secondary alignment. The system digitizes a workflow that is often still manual, introduces AI-assisted decision support, and connects forecasting, replenishment, and risk detection into a single operational flow. In that sense, the solution is not only about reducing waste, but also about modernizing bakery operations in a practical, explainable, and scalable way.

Problem Statement

Bakery businesses, from single-shop operators to multi-outlet bakery-cafe chains, do not only need visibility into what was sold or what stock remains. They need a reliable answer to a more urgent operational question:

How many units of each bakery item should a bakery prepare or receive tomorrow, whether for one shop or for multiple outlets, and what should the kitchen and purchasing team do today to support that plan?

Existing restaurant and retail systems often provide transaction history, item availability, inventory counts, or procurement records. However, these tools do not typically convert that information into day-ahead shop-level or outlet-by-outlet, daypart-aware prep decisions for short-shelf-life bakery products. As a result, operators still rely heavily on manual judgment, which can be slow, inconsistent, and vulnerable to overreaction or underreaction.

For bakery businesses, whether a single shop or a multi-outlet chain, this gap has direct consequences:

- overproduction and end-of-day waste
- stockouts during morning and lunch peaks

- poor central-kitchen allocation across outlets
- unnecessary ingredient spend
- wasted labor and production capacity
- weaker consistency in freshness and customer experience

Objectives

The objectives of Predictory are:

- reduce overproduction and end-of-day bakery waste
- reduce stockouts during peak selling windows
- improve shop-level and outlet-level prep decisions by date and daypart
- translate prep plans into ingredient replenishment actions
- provide explainable recommendations that users can review and override
- reduce planning time and increase planning consistency across one shop or many outlets
- give kitchen and operations teams a clearer, action-oriented view of tomorrow's plan

Review of Existing Solutions

Current market solutions fall into three broad categories: POS/reporting platforms, inventory and procurement tools, and general demand-planning systems. Predictory is intended to sit between those categories by focusing specifically on next-day bakery action support.

Competitor and Existing Solution Analysis

Category	Example	What It Does Well	Gap Relative to Predictory
POS + restaurant inventory	Square Restaurant Inventory by MarketMan	Ingredient-level tracking, waste documentation, par levels, purchase orders, invoice scanning, multi-location visibility, POS sync [4]	Strong at tracking and control, but not centered on bakery-specific shop/daypart or outlet/daypart prep recommendations for tomorrow
Restaurant inventory and finance	Toast xtraCHEF / Toast Inventory Management	Invoice automation, recipe costing, inventory management, vendor and product catalog centralization, margin visibility [5][6]	Strong back-office control, but less focused on operational prep planning as a single bakery workflow
Inventory and procurement platform	MarketMan	Purchasing, supplier management, food costing, multi-location HQ management, POS and distributor integrations [7]	Strong procurement and cost management, but less specialized for bakery demand-by-daypart decision support
Restaurant operations and commissary	Restaurant365	Inventory, purchasing, commissary, multi-location reporting, waste reduction, production and fulfillment workflows [8][9]	Covers broader restaurant operations well, but is not optimized around the specific day-ahead bakery planning ritual

Category	Example	What It Does Well	Gap Relative to Predictory
Enterprise demand planning	Anaplan Demand Planning	Collaborative planning, driver-based demand planning, scenario analysis, enterprise coordination [10]	Powerful but broader and more enterprise-oriented than the needs of small-to-mid-sized bakery businesses, whether single-shop or multi-outlet
F&B Procurement & Inventory	Food Market Hub	Automated procurement, real-time inventory tracking, daily waste recording, AI-powered sales forecasting to optimize purchasing [11]	Strong on procurement and supply chain, but lacks specialized daypart-aware bakery production (prep) plan generation and AI-assisted causality explanations

Gaps Predictory Intends to Fix

Predictory is designed to close a specific gap between reporting tools and action tools. It does this by:

- forecasting demand by outlet and daypart
- converting forecast into prep recommendations
- converting prep into ingredient replenishment needs
- flagging likely waste and stockout risks before production
- providing plain-language explanations and scenario simulation
- keeping a human-in-the-loop edit and approval workflow

In short, existing tools are often strongest at recording, tracking, costing, and reporting. Predictory aims to be strongest at **next-day bakery planning and explainable action support**.

Methodology

Technical Team Focus

Predictory is implemented as a modular monolith with a modern web frontend, a Python backend, a relational database layer, and an AI layer used for explanation and scenario support. The current repository uses:

- **Frontend:** Next.js 14, React 18, TypeScript, Tailwind CSS, TanStack Query, Zustand, Recharts, Lucide
- **Backend:** FastAPI, Uvicorn, SQLAlchemy 2.x, Alembic, Pydantic v2
- **Data and analytics libraries:** Pandas, NumPy, python-dateutil
- **AI layer:** LiteLLM and LangGraph
- **Database:** PostgreSQL in the intended architecture, with local SQLite support for demo use

The implementation approach combines deterministic planning logic with AI-assisted explanation. This is important because the prototype is intended to be operationally credible, not a black-box model demo. The forecasting and planning layers use explicit formulas and heuristics, while the AI layer is used for explanations, summarization, action phrasing, and scenario interpretation.

Prototype-Accurate Architecture Diagram

```
                +---------------------------+
                |      User / Browser       |
                |   Dashboard + Planning    |
                +-------------+-------------+
                              |
                              | HTTP
                              v
                +-------------+-------------+
                |   Next.js Frontend (web)  |
                | dashboard, forecast, prep |
                | replenishment, risk, AI   |
                +-------------+-------------+
                              |
                              | REST API
                              v
        +----------------------+----------------------+
        |           FastAPI Backend (api)             |
        | catalog | ingestion | forecasting | copilot|
        | planning | alerts | ops_data                |
        +-----------+-----------+-----------+--------+
                    |           |           |
                    |           |           +------------------+
                    |           |                              |
                    v           v                              v
          +---------+--+  +----+----------------+   +----------+---------+
          | Forecasting |  | Planning + Alerts   |   | Copilot / AI Layer |
          | baseline    |  | prep, replenishment,|   | explain, brief,    |
          | demand +    |  | waste, stockout     |   | actions, scenario  |
          | drivers     |  | logic               |   | via LiteLLM        |
          +----+--------+  +----+----------------+   +----------+---------+
               |               |                              |
               +---------------+--------------+---------------+
                                              |
                                              v
                                  +-----------+-----------+
                                  | PostgreSQL / SQLite   |
                                  | sales, stock, waste,  |
                                  | plans, overrides,     |
                                  | holidays, copilot     |
                                  +-----------+-----------+
                                              |
                    +---------------------------+---------------------------+
                    |                           |                           |
                    v                           v                           v
          +---------+----------+      +----------+----------+      +---------+----------+
          | Seeded Demo Data   |      | CSV Imports         |      | Optional Weather   |
          | outlets, SKUs,     |      | sales, inventory,   |      | signal (Open-Meteo)|
          | waste, stockouts   |      | products, holidays  |      | for forecast layer |
          +--------------------+      +---------------------+      +--------------------+
```

Technical Workflow

The implemented prototype workflow is:

1. seed or ingest operational data
2. forecast demand by outlet, SKU, and daypart
3. convert forecasts into prep recommendations
4. convert prep requirements into ingredient replenishment

5. compute waste and stockout risks
6. present the results in a decision-oriented web UI
7. layer AI explanation, daily briefs, daily actions, and scenarios on top

Forecasting Logic

The repository implements a baseline demand forecasting engine using:

- weighted recent sales
- same-weekday pattern
- 14-day moving average
- historical daypart split ratios

The forecast layer has also been expanded with contextual demand drivers:

- holiday flags
- weather signals
- manual event and promo overrides
- a simple stockout-censoring heuristic to recover likely lost sales when inventory history suggests demand was capped by stockouts

This makes the forecasting engine more convincing than a simple moving average, while remaining explainable and practical for a prototype.

Planning and Risk Logic

The planning engine translates forecasts into actions for a single bakery shop or a network of outlets:

- prep recommendations by outlet, SKU, and daypart
- ingredient replenishment needs from recipes/BOM
- waste and stockout alerts based on operational thresholds

The current implementation also supports:

- manual prep edits
- prep approval workflow
- forecast context inspection
- forecast overrides
- daily planning summary generation

Workflow and Interface Decisions

Although this is a technical submission, the system still required clear workflow design. The repository does not contain evidence of formal field interviews, pilot validation, or structured usability studies with bakery operators. Therefore, this section is presented only as implementation-supporting context rather than as a separate non-technical research track.

Problem Framing

The team began by identifying a domain-specific pain point: bakery businesses face high uncertainty in next-day production, whether they run a single shop or several outlets, especially when products are perishable

and demand differs by daypart. The product requirement document clearly frames the solution around this decision problem rather than around generic inventory or transaction tracking.

Operator Roles Reflected in the System

The prototype implies the following user roles:

- owner or founder-operator
- operations manager
- kitchen or central kitchen manager
- outlet manager
- purchaser or inventory lead

These roles are grounded in the operating model described in the PRD and reflected directly in the planning, approval, replenishment, and explanation flows built into the application.

Workflow Design

The screens are designed around a practical evening planning ritual for either a single bakery shop or a multi-outlet operation:

1. open the dashboard for tomorrow's operational overview
2. inspect the forecast and demand drivers
3. edit or approve prep plans
4. review ingredient replenishment
5. inspect waste and stockout risks
6. use AI explanation and scenarios to support decisions

This is an important implementation decision because it makes the interface decision-oriented rather than report-oriented, which is critical for a planning system intended for daily operational use.

Key Features and Explanation

1. Executive Dashboard

The dashboard consolidates predicted sales, risk scores, at-risk shops or outlets, top deterministic actions, and optional AI action plans. It gives decision-makers a quick overview before they move into detailed operational pages.

2. Forecasting With Demand Drivers

The forecast page does more than show baseline demand. It surfaces:

- shop/daypart or outlet/daypart forecast lines
- holiday flags
- weather signals
- manual event and promo overrides
- stockout-recovery context

This is important because a planning tool becomes more trustworthy when users can see not only the forecast number, but also the demand drivers behind it.

3. Editable Prep Planning

The prep-plan flow turns forecast demand into recommended prep units by shop or outlet, SKU, and daypart. Users can edit lines and approve the plan. This human-in-the-loop design is important because frontline teams usually want support, not blind automation.

4. Ingredient Replenishment

The replenishment module converts prep requirements into ingredient needs using recipe/BOM data. This makes the system operationally coherent: forecast leads to prep, and prep leads to ingredient action.

5. Waste and Stockout Risk Center

The risk-center shows waste and stockout alerts before service begins, rather than only reporting them after the fact. This moves the product from passive reporting toward proactive intervention.

6. AI Explainability and Copilot Support

The copilot layer provides:

- explain-plan responses
- daily brief generation
- structured daily actions
- scenario interpretation

This makes the system easier to understand and defend operationally, especially for teams that need narrative summaries rather than raw tables.

7. Multilingual Experience

The prototype supports:

- English
- Bahasa Melayu
- Simplified Chinese

This adds inclusivity and practical usability for ASEAN-facing teams with mixed language preferences.

Tech Stacks Used

Frontend Stack

- Next.js 14
- React 18
- TypeScript
- Tailwind CSS
- TanStack Query
- Zustand

- Recharts
- Lucide React

Backend Stack

- FastAPI
- Uvicorn
- SQLAlchemy 2.x
- Alembic
- Pydantic v2
- python-dotenv
- python-multipart
- httpx

Data and Computation

- PostgreSQL or local SQLite
- Pandas
- NumPy
- python-dateutil

AI and Agent Tooling

- LiteLLM
- LangGraph
- Gemini support through LiteLLM-compatible provider configuration

Implementations, Innovations, and Business Value

What Was Implemented

Based on the repository, the prototype currently implements:

- ingestion APIs for products, sales, inventory, and holidays
- catalog APIs for outlets, SKUs, ingredients, and recipes
- forecast generation
- forecast context and override management
- editable prep planning with approval
- replenishment plan generation
- waste and stockout alert generation
- AI explain-plan, daily brief, daily actions, and scenario APIs
- multilingual frontend and copilot support

Technical Contribution Summary

From a technical-team perspective, the main contribution of Predictory is not a single algorithm but the integration of multiple operational modules into one coherent system:

- a FastAPI backend with clear module boundaries for forecasting, planning, alerts, ingestion, and copilot features

- a frontend that consumes those contracts through typed integrations rather than static mock data
- a deterministic decision engine that remains auditable and explainable
- an AI layer that is intentionally constrained to explanation, summarization, action phrasing, and scenario support
- a seeded dataset engineered to demonstrate realistic bakery failure modes such as waste hotspots and morning stockouts

This combination is what makes the prototype technically credible for a hackathon setting.

What Is Innovative in This Prototype

The prototype is innovative not because it claims to be a custom-trained ML system, but because it combines several layers into a single operational flow:

- deterministic demand forecasting
- daypart-aware bakery planning
- ingredient-level replenishment
- proactive waste and stockout risk surfacing
- human-in-the-loop override and approval
- explainable AI on top of operational logic
- multilingual accessibility

The result is more credible than a pure dashboard, and more controllable than a fully autonomous AI workflow.

Business Value

If validated in real bakery operations, the business value of Predictory would likely come from:

- lower end-of-day waste
- fewer missed sales from stockouts
- better alignment between the kitchen and the point of sale, whether one shop or many outlets
- reduced over-ordering of ingredients
- faster and more consistent daily planning
- clearer reasoning for operational decisions

The prototype should not yet claim live ROI, but it does demonstrate a clear path to operational and financial value.

Results

Prototype Outcome

Predictory delivers a working prototype across the core operational workflow it set out to solve. Based on the current repository, the project includes:

- demand forecasting by shop or outlet and daypart
- prep plan generation
- ingredient replenishment planning
- waste and stockout alerts

- AI explanation features
- daily action planning
- scenario simulation
- multilingual UI and copilot support

The prototype is not just a collection of isolated screens. It is an end-to-end planning flow in which forecast outputs feed prep logic, prep feeds replenishment, and both forecast and plan states drive risk detection and copilot outputs.

Evidence From the Repository

The implementation is backed by concrete engineering artifacts:

- backend contracts for ingestion, planning, alerts, and copilot
- seeded demo data representing five outlets and multiple SKUs while still reflecting logic that also applies to a single-shop bakery
- explicit demo scenarios for Bangsar waste risk and Mid Valley stockout risk
- frontend pages for dashboard, forecast, prep plan, replenishment, risk center, copilot, and scenario planning
- automated backend test coverage across forecasting, prep, replenishment, alerting, data APIs, forecast context, and copilot behavior

Engineering Validation

From a technical-team perspective, the strongest evidence in this prototype is not only that screens exist, but that the modules are wired together coherently and validated in code:

- backend API contracts are documented in `apps/api/CONTRACTS.md`
- seeded data supports repeatable demo scenarios rather than random mock values
- forecasting, prep, replenishment, alerts, and copilot modules are separated in the backend codebase
- backend tests cover forecasting, prep, replenishment, ingestion, alerts, forecast context, and copilot flows
- frontend integration work was validated through successful typecheck and lint runs
- the remaining local build issue was environment-specific (`spawn EPERM` under Windows/OneDrive), not a known application logic failure

In the latest verified development cycle before this report, the project had a passing backend test suite and passing frontend typecheck and lint. The remaining build issue documented during development was an environment-specific local Next.js worker `spawn EPERM` problem, not a known product logic or typing failure.

Objectives Met

The prototype substantially meets its core objectives at the prototype level:

Objective: Reduce overproduction and waste
Met at the prototype decision-support level through waste alerts, prep recommendations, and risk scoring.

Objective: Reduce stockouts
Met at the prototype decision-support level through demand forecasting, stockout alerts, and replenishment logic.

Objective: Improve outlet-level prep decisions
Met through shop-level or outlet/daypart forecasting and editable prep planning.

Objective: Convert prep into replenishment action
Met through BOM-driven ingredient recommendations and urgency classification.

Objective: Provide explainable recommendations
Met through AI explanations, daily briefs, and scenario interpretation.

Objective: Keep humans in control
Met through manual edits, override logic, and approval flows.

What Is Not Yet Proven

The system should not yet claim real-world impact metrics such as:

- measured waste reduction in live bakery operations
- measured stockout reduction in live customer environments
- validated forecast accuracy on production POS data
- full operational ROI after deployment

These outcomes remain future validation targets, not proven prototype results. The correct claim is that Predictory demonstrates a credible, explainable operational planning prototype with realistic seeded evidence and a clear path toward measurable business impact.

Quantitative Development Evidence

The prototype includes several concrete indicators of engineering completeness:

- 5 demo outlets
- 8 seeded SKUs
- 30 days of seeded history
- 7 major user-facing routes
- multilingual UI and copilot outputs in 3 languages
- seeded holidays and outlet coordinates
- automated backend test coverage across multiple functional domains

Simulated Demo Results

To provide quantitative evidence without overclaiming real-world outcomes, the following results are derived directly from Predictory's seeded operational dataset, which uses realistic operational patterns to replicate typical bakery failure modes.

Demo Dataset Overview

Metric	Value
Outlets simulated	3 (KLCC, Bangsar, Mid Valley)
SKUs tracked	5 (Butter Croissant, Chocolate Muffin, Banana Bread, Cheese Danish, Cinnamon Roll)

Metric	Value
Historical sales window	30 days
Ingredients tracked via BOM	8
Holidays seeded	6 (including CNY +35% uplift, Christmas +25%, New Year +15%)
Weekend demand multiplier	×1.25 above base

Simulated Operational Patterns Detected

Scenario	Outlet	Pattern	How Predictory Surfaced It
Chronic overproduction	Roti Lane Bangsar	Butter Croissant prepped **~18% above actual sales** for 30 consecutive days → **~15% end-of-day waste rate**	Waste alert generated; prep plan recommendation reduced by approx. 15 units/day
High-frequency stockout	Roti Lane Mid Valley	Morning Butter Croissant sold out on **4 of every 7 days** (Mon/Wed/Fri/Sat)	Stockout alert generated; AI brief flagged lost revenue opportunity and suggested morning batch increase
Demand spike on holidays	All outlets	CNY holiday applied **+35% demand uplift** across all SKUs	Forecast layer auto-adjusted quantities; Demand Drivers panel displayed the uplift flag
Weekend surge	All outlets	Demand runs **25% higher** on Saturdays and Sundays	Forecast engine detected weekday vs. weekend pattern and adjusted prep recommendations accordingly

Forecasted Daily Demand (Base Weekday, All Outlets Combined)

SKU	KLCC	Bangsar	Mid Valley	Total/day
Butter Croissant	45	35	50	**130 units**
Cheese Danish	25	20	30	**75 units**
Chocolate Muffin	30	25	35	**90 units**
Cinnamon Roll	20	16	22	**58 units**
Banana Bread	15	12	18	**45 units**
Total	**135**	**108**	**155**	**≈ 398 units/day**

*(Saturday/Sunday projected at ×1.25, giving approximately **498 units/day** across 3 outlets on weekends.)*

Risk Alerts Generated (Simulated 30-Day Run)

Alert Type	Outlet	Product	Detection Method
Waste Hotspot	Roti Lane Bangsar	Butter Croissant	Waste rate exceeded 15% threshold for 30 consecutive days
Stockout Risk	Roti Lane Mid Valley	Butter Croissant	Morning EOD inventory reached 0 on 4 of 7 days per week
Demand Driver: Holiday	All outlets	All SKUs	Upcoming CNY (+35%) and Demo Festival Day (+5%) detected
Demand Driver: Weekend	All outlets	All SKUs	Weekend multiplier ×1.25 automatically applied to forecast

Key System Outcomes

These simulated results demonstrate three core operational improvements that Predictory is designed to deliver:

1. **Earlier waste detection.** The Bangsar overproduction pattern (15%+ waste rate on Croissants) was identified on **day 1 of the seeded window**, not after 30 days of manual observation. In a real deployment, this could translate to preventing weeks of cumulative loss before human observation catches it.

2. **Proactive stockout prevention.** The Mid Valley morning stockout pattern (4× per week) was flagged as a recurring risk, prompting a targeted prep recommendation increase. With a 25% uplift to the morning batch (~6 additional units), estimated recoverable revenue per day could reach **RM 51 (6 units × RM 8.50)**, or approximately **RM 1,428/month** for this single outlet-SKU combination.

3. **Context-aware forecasting.** Demand driver signals (holidays, weekends) adjusted the baseline automatically, demonstrating that the forecasting layer can incorporate contextual knowledge without requiring manual recalibration by the bakery team.

> **Disclaimer:** These figures are derived from the prototype's seeded simulation environment. They illustrate the system's detection capability and directional business impact under controlled demo conditions, not live operational data. Real-world results will depend on actual POS, inventory, and operations data from deployed bakery environments.

Prototype Showcase

While this is a technical submission, interface evidence is still useful because it demonstrates that the implemented backend modules are exposed through coherent operator workflows. This section uses real captured screens from the working prototype.

Screenshot 1: Executive Overview Dashboard

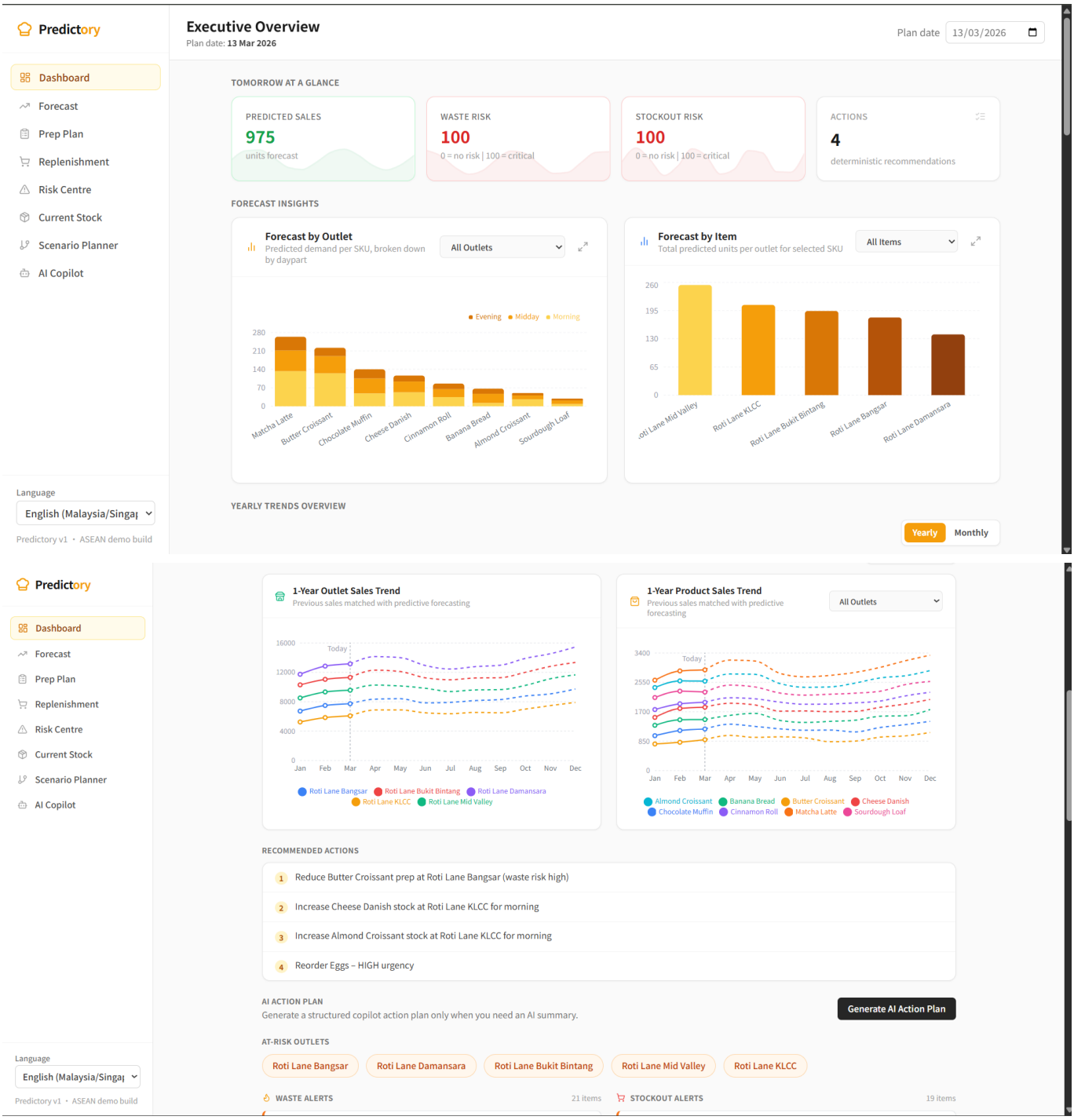

Figure 1. Executive Overview dashboard. This screen consolidates predicted sales, waste risk, stockout risk, recommended actions, and at-risk shops or outlets into a single planning view. It matters because operators need a fast decision summary before reviewing detailed plans.

Screenshot 2: Forecast and Demand Drivers

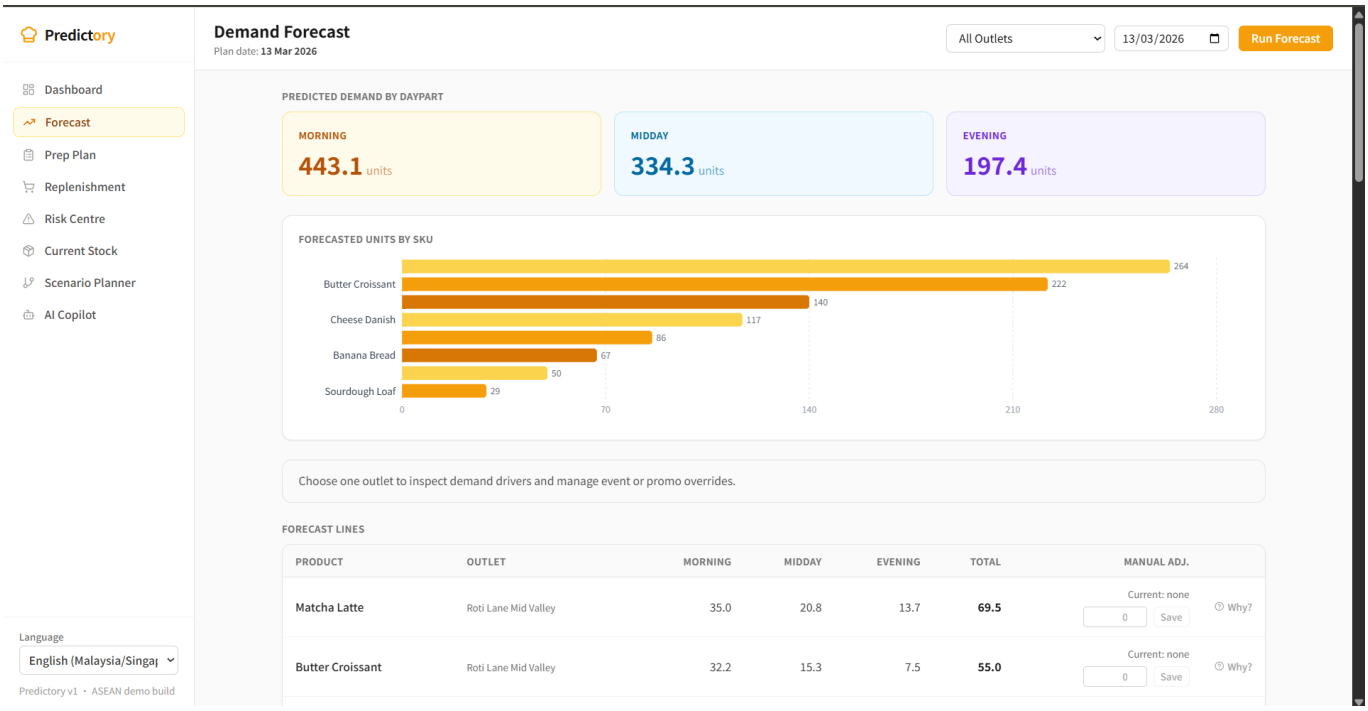

Figure 2. Forecast screen with context-aware demand drivers. The forecast view combines shop/daypart or outlet/daypart demand lines with holiday, weather, override, and stockout-recovery context. It matters because bakery planning depends on more than raw historical averages.

Screenshot 3: Prep Plan and Approval Flow

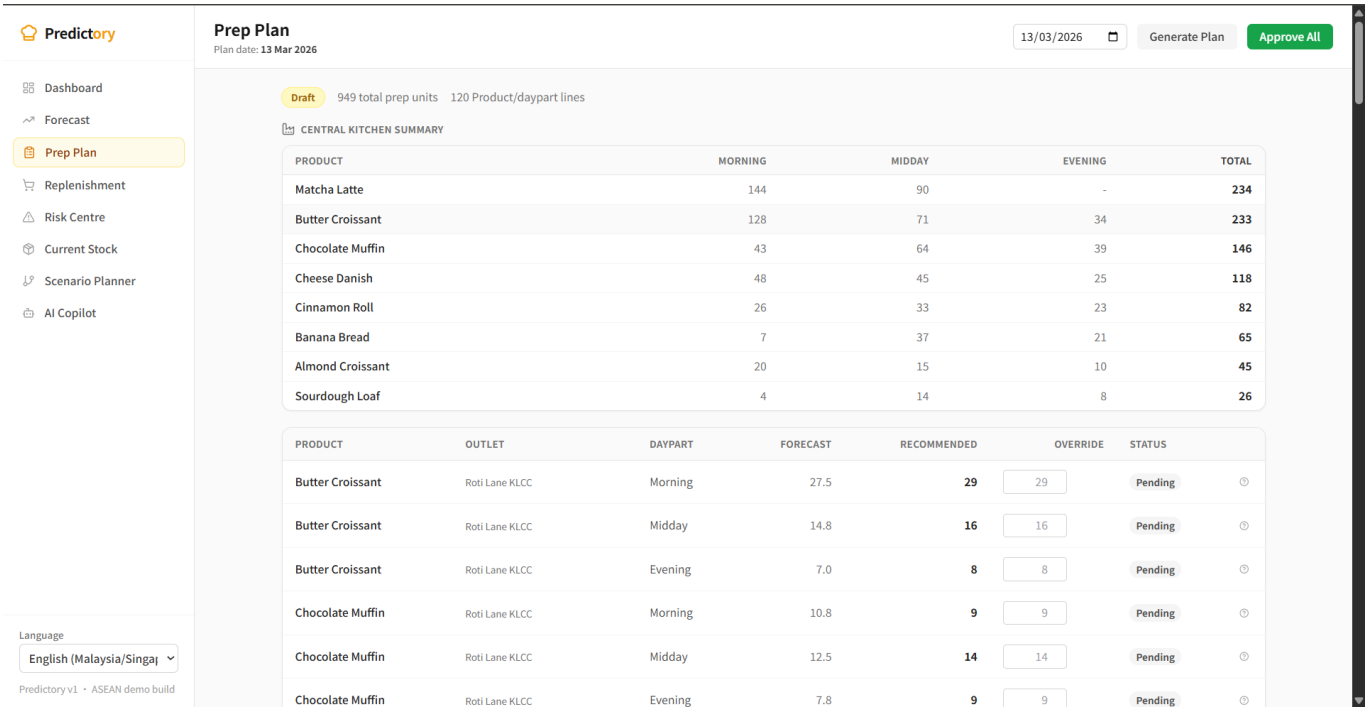

Figure 3. Prep planning with human-in-the-loop editing. Users can inspect recommended prep quantities, adjust line items, and approve the plan for execution. It matters because operational trust requires editable recommendations, not fully automated decisions.

Screenshot 4: Ingredient Replenishment

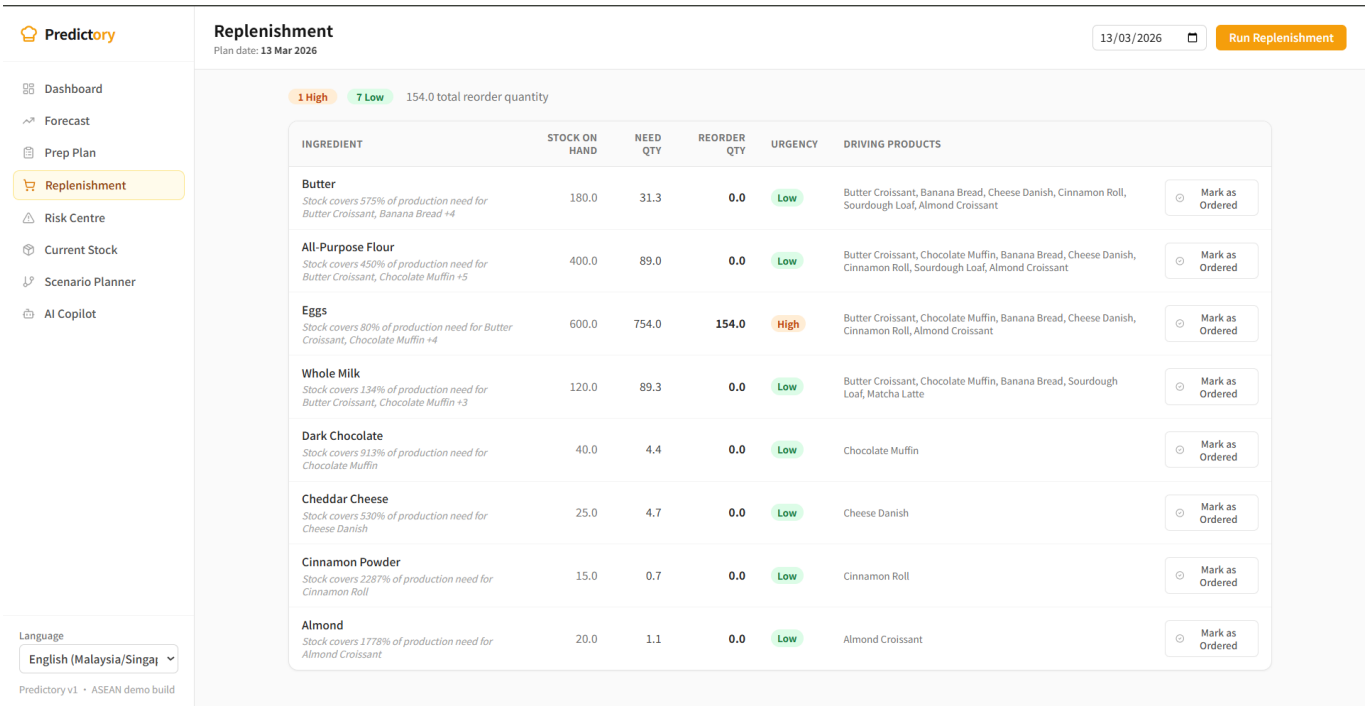

Figure 4. Ingredient replenishment planning. The system converts SKU-level prep needs into ingredient reorder actions with urgency signals. It matters because production planning is only useful if ingredient availability is also managed.

Screenshot 5: Risk Centre

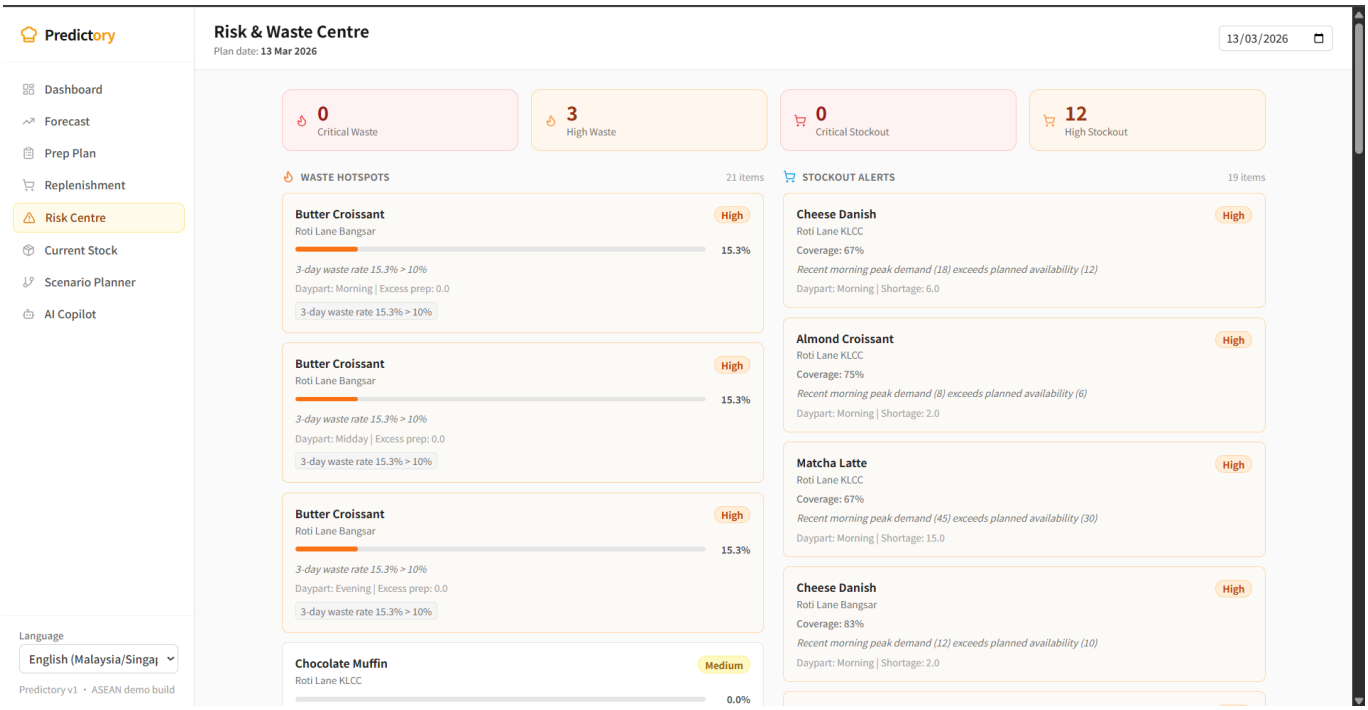

Figure 5. Risk centre for proactive issue detection. Waste and stockout alerts are surfaced before service begins, allowing operators to intervene earlier. It matters because preventing problems is more valuable than only reporting them after the fact.

Screenshot 6: AI Copilot

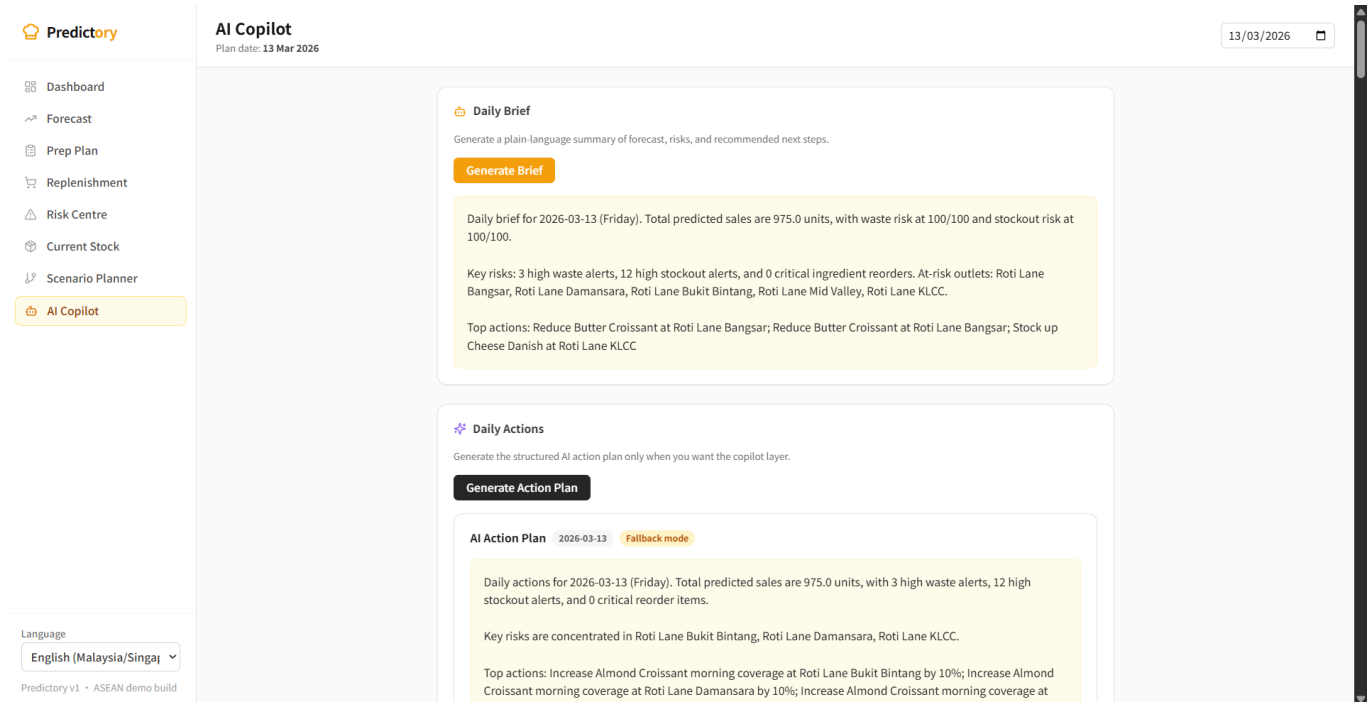

Figure 6. AI copilot for explainable action support. The copilot layer converts operational signals into readable daily briefs and structured action recommendations. It matters because decision support is more useful when teams can quickly understand the reasoning behind a plan.

Screenshot 7: Scenario Planner

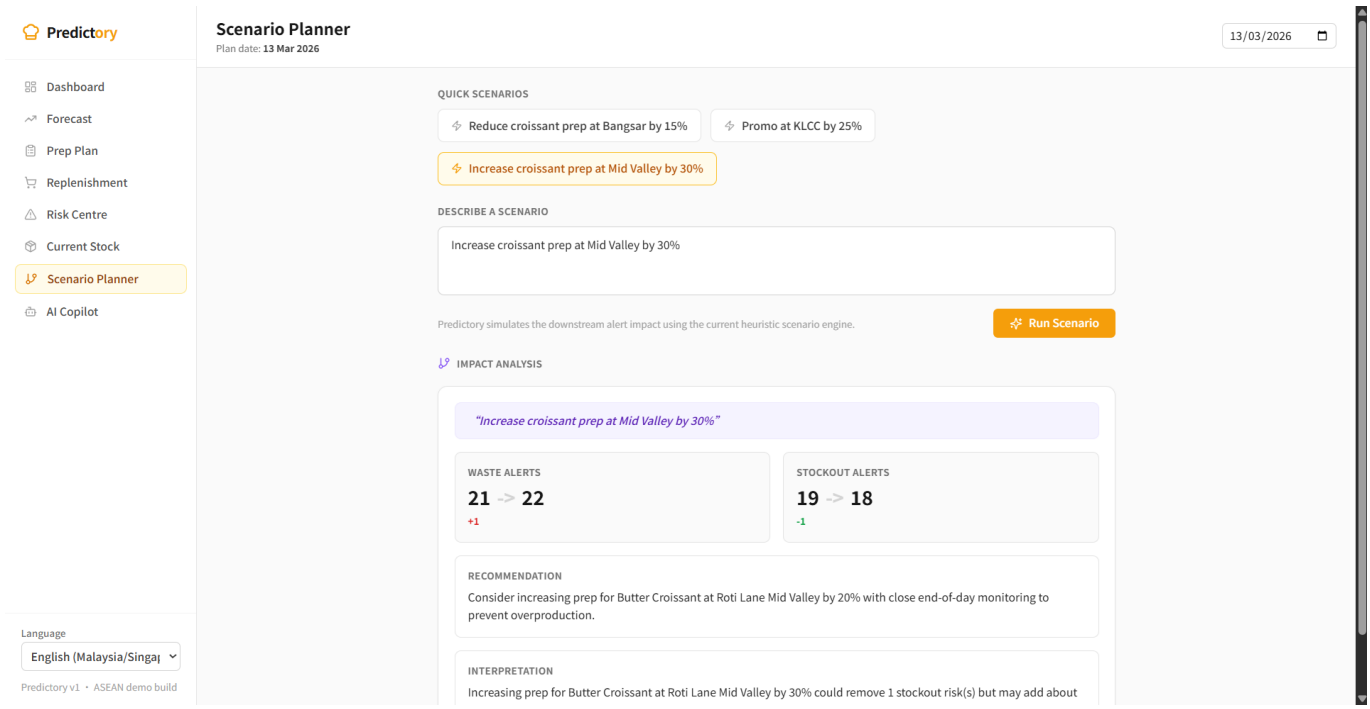

Figure 7. Scenario planning interface. Users can test hypothetical operational changes and inspect how waste and stockout risks may shift. It matters because safe experimentation supports better planning under uncertainty.

Screenshot 8: Current Stock Inventory

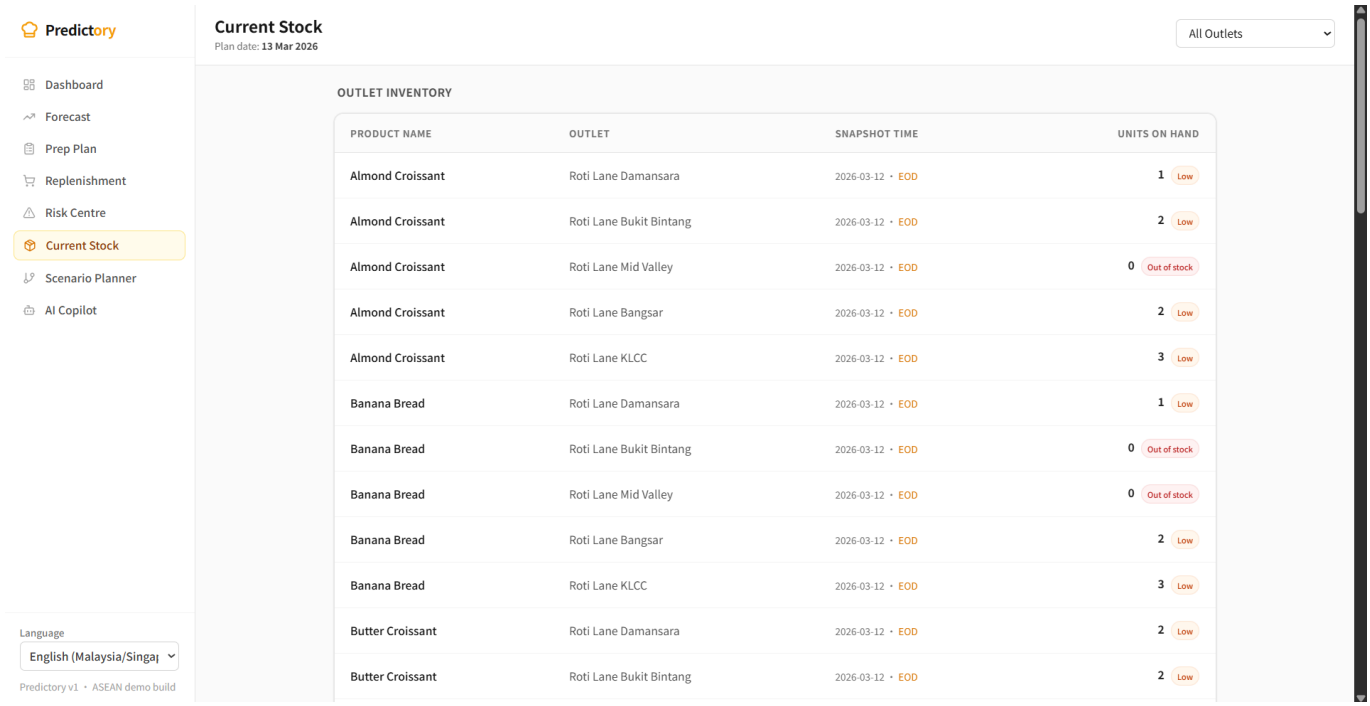

Figure 8. Current Stock inventory monitoring. This view provides a unified snapshot of stock levels for all products across multiple outlet locations. It matters because real-time stock visibility is the foundation for accurate replenishment and waste prevention.

Screenshot 9: Multilingual Experience

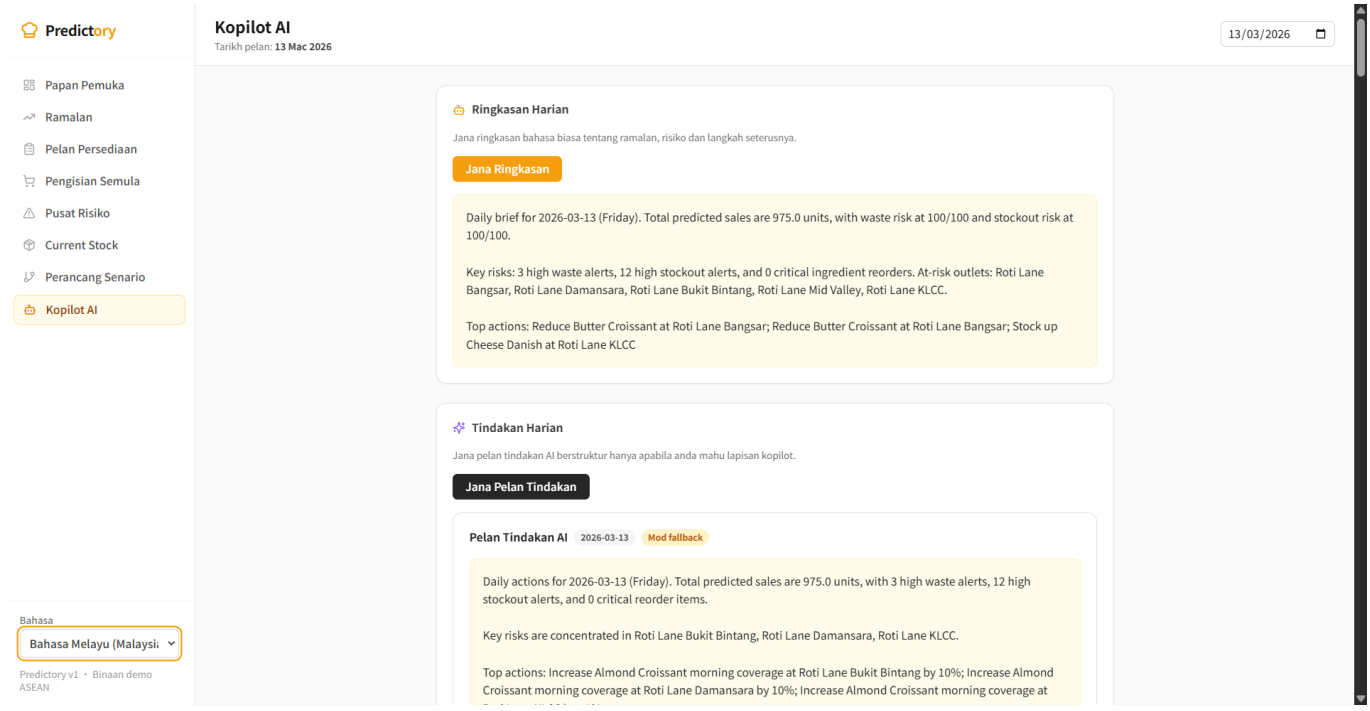

Figure 9. Multilingual UI for inclusive adoption. Predictory supports English, Bahasa Melayu, and Simplified Chinese in the interface and copilot outputs. It matters because accessibility and usability improve when teams can interact with the system in familiar languages.

Challenges Faced

The development process revealed several practical challenges:

1. Data realism versus hackathon time

A convincing bakery-planning demo needs more than random sample data. The team had to create seeded data that reflected realistic operational conditions such as shop-specific or outlet-specific demand, daypart behavior, waste patterns, and stockout patterns.

2. Avoiding overclaiming on AI and ML

A major challenge was balancing credibility with ambition. The product needed to feel intelligent without pretending to have a production-grade trained ML model or validated real-world business impact. The solution was to keep business-critical planning logic deterministic and use AI mainly for explanation, summarization, and scenario support.

3. Frontend and backend contract drift

As the backend became more complete, the frontend had to be realigned to the actual API contracts. This required freezing backend response shapes and then updating types, request payloads, and screen assumptions so the web application matched real backend behavior rather than stale local assumptions.

4. Demo-safe integration of demand drivers

Adding holiday, weather, overrides, and stockout-censoring logic improved realism, but also increased the risk of confusing users if the drivers were hidden. The challenge was solved by exposing a forecast-context layer and a demand-drivers panel in the UI.

5. Environment-specific local build issues

During development, the frontend encountered a local Next.js `spawn EPERM` issue in the Windows/OneDrive environment. This did not invalidate the app logic, but it highlighted an engineering challenge around local build environments and reproducibility.

6. Multilingual support without over-engineering

The app needed to support English, Bahasa Melayu, and Simplified Chinese without introducing a heavy routing-based i18n architecture. The pragmatic solution was an in-app language provider with localized UI strings and localized copilot prose.

Detailed Future Roadmap

Phase 1: Pilot Readiness

- deploy the current prototype to a stable hosted environment
- run pilot onboarding with one real bakery operator, whether a single shop or a multi-outlet business
- validate forecast quality using real POS and inventory exports
- measure planning time saved, waste trend changes, and stockout reduction trends
- harden authentication, audit logging, and operational monitoring
- add CI-backed regression checks for critical forecast, prep, alert, and copilot flows

Phase 2: Operational Data Integration

- connect to live POS systems
- connect to inventory and purchasing systems
- automate ingestion of sales, item catalog, shop or outlet master data, inventory, and promotions
- keep CSV import as fallback rather than the primary workflow
- add connector abstractions so vendor-specific integrations do not leak into core planning logic

Phase 3: Smarter Demand Drivers

- strengthen weather logic with more outlet-specific signal tuning
- add richer holiday calendars and region-specific events
- add manual and scheduled promo calendars
- improve stockout-censoring and lost-sales estimation

Phase 4: ML Model Integration

The current prototype uses a deterministic, heuristic-based forecasting engine (weighted moving average + weekday patterns + demand driver multipliers). This is intentional — it allows explainable outputs, fast deployment, and zero training data requirements. However, as real operational data accumulates from pilot bakeries, the system is designed to progressively incorporate machine learning models.

The proposed ML integration roadmap is as follows:

Stage 4a: Baseline Benchmarking

Before introducing any ML model, validate that the deterministic baseline is credible on real held-out data:

- split 30–90 days of real POS data into train/test windows
- compute MAE, RMSE, and MAPE on daily SKU-outlet demand forecasts
- use this as the benchmark that any ML model must beat to be deployed

Stage 4b: Gradient Boosting for Short-Term Demand Forecasting

- **Model:** XGBoost or LightGBM trained per SKU or per SKU-outlet combination
- **Features:** weekday, daypart, lag sales (1-day, 7-day, 14-day), rolling mean, holiday flag, weather signal, promo flag
- **Training cadence:** weekly retrain on rolling window of recent data
- **Deployment:** model artifacts stored as versioned pickle or ONNX files, called via FastAPI inference endpoints
- **Benefit:** captures non-linear interactions between demand drivers (e.g., rain + weekend + promo at the same time) that a simple multiplier chain cannot model

Stage 4c: Time-Series Foundation Models

For bakeries with sufficient data history (12+ months), evaluate pretrained time-series models:

- **Models:** NeuralForecast (NHITS, PatchTST), or TimesFM (Google)
- **Benefits:** strong zero-shot generalization for new outlet-SKU combinations, without needing per-SKU training data
- **Use case:** ideal for expanding to new outlets where historical sales data is limited

Stage 4d: Waste and Stockout Prediction Models

- **Task:** binary classification — will a given outlet-SKU combination result in waste or stockout tomorrow?
- **Model:** Logistic Regression → Random Forest → Gradient Boosting (progressively as data grows)
- **Input features:** forecast delta (predicted vs. recent average), stock on hand, days since last stockout, daypart split ratio, weather, holiday proximity
- **Output:** risk probability score (0–100), which replaces the current threshold-based alert logic
- **Benefit:** alert quality improves over time as the model learns from real outcomes

Stage 4e: Promo Elasticity and SKU Substitution

- estimate demand uplift per promotional event type using historical promo tagging
- model cross-SKU substitution effects (e.g., if Butter Croissant sells out, does Cheese Danish sales lift?)
- use regression or causal inference techniques once enough promo-tagged data is available

ML Governance Principles

Predictory's ML integration will follow strict human-in-the-loop governance:

- no ML model output replaces a human decision without an override path
- all model predictions are surfaced with confidence signals and explainability context
- deterministic fallback is always maintained as a safety net if model confidence is low
- model performance is monitored continuously; degradation triggers automatic fallback

Phase 5: Product and Business Expansion

- improve multi-organization support
- expand role-based workflows for head office teams, outlet teams, and lean single-shop teams
- add richer executive analytics and trend reporting
- build a stronger onboarding flow for CSV mapping and account setup
- add notifications and monitoring for planning exceptions
- improve deployment hardening, observability, and runtime performance for real pilot usage

AI Acknowledgement (Mandatory)

Disclosure

This project utilized a multi-layered AI stack for research, development, and as a core part of the application’s production intelligence.

AI Category	AI Tool	How It Was Used
Production Intelligence	Google Gemini 2.5 (API via LiteLLM)	Integrated into the backend to generate Daily Forecast Briefs, prioritized Daily Actions, and plain-language reasoning for production decisions.
Development Support	Google Gemini (Web), Antigravity (Agentic Helper)	Used for overall project orchestration, UI design implementation, agent-to-environment interaction, and terminal automation.

AI Category	AI Tool	How It Was Used
Coding & Refinement	GitHub Copilot, OpenAI Codex	Used for code generation, boilerplate reduction, unit test writing, and complex logic debugging across React/FastAPI.
Deep Research	Perplexity AI, ChatGPT (GPT-5.4 Pro)	Used for market analysis (competitor research), SDG alignment verification, and statistical data gathering for Malaysia's food waste.
Asset Generation	Google Nano Banana 2.0, NotebookLM	Used to generate professional infographics, tech stack visuals, and slide assets for the pitch deck and report.
Collaboration	Notion AI	Used for team collaboration, documentation drafting, and meeting note summarization during the ideation phase.

Conclusion

Predictory is a focused, operationally grounded prototype that addresses a real problem in bakery operations, from single-shop bakeries to bakery-cafe chains: how to plan tomorrow's prep and replenishment decisions more accurately than manual judgment or historical averages alone. Instead of attempting to replace POS or ERP systems, it acts as a decision-support layer that converts operational data into shop/daypart or outlet/daypart forecasts, prep plans, replenishment needs, and proactive risk signals.

The project aligns most strongly with **SDG 12** by targeting waste reduction and more responsible production planning for perishable goods. It also supports **SDG 9** by digitizing and modernizing a planning workflow that is often still manual. The prototype demonstrates meaningful progress through an end-to-end workflow, seeded operational evidence, explainable AI assistance, and multilingual accessibility.

The next stage of development should focus on:

- live POS and inventory integrations
- pilot deployment with real bakery operators
- stronger validation of forecast quality and business impact
- additional demand drivers such as promotions, events, and richer weather logic
- advanced modeling such as elasticity, substitution, and deeper lost-sales estimation

Overall, Predictory is a strong prototype because it is not merely a concept or dashboard mockup. It is a working decision-support system with a clear problem focus, a credible technical foundation, and a sustainability narrative that is specific, defensible, and relevant to real bakery operations.

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